Software Requirements Specification

for

NutriSafe AI: A Multi-Modal Explainable AI System for Personalized Pediatric Food Additive Risk and Chemical Interaction Assessment

Version 1.0

Prepared by P.K.R. Abeysekara (E2529001)

Faculty of Information Technology, University of Moratuwa

26 September 2026

Table of Contents

1. Introduction 1
1.1 Purpose 1
1.2 Document Conventions 1
1.3 Intended Audience and Reading Suggestions 1
1.4 Product Scope 2
1.5 References 2
2. Overall Description 4
2.1 Product Perspective 4
2.2 Product Functions 5
2.3 User Classes and Characteristics 6
2.4 Operating Environment 7
2.5 Design and Implementation Constraints 7
2.6 User Documentation 8
2.7 Assumptions and Dependencies 8
3. External Interface Requirements 9
3.1 User Interfaces 9
3.2 Hardware Interfaces 12
3.3 Software Interfaces 12
3.4 Communications Interfaces 13
4. System Features 14
4.1 Food Label, QR & Barcode Scanning 14
4.1.1 Description and Priority 14
4.1.2 Stimulus/Response Sequences 14
4.1.3 Functional Requirements 15
4.2 Child Health Profile Management 15
4.2.1 Description and Priority 15
4.2.2 Stimulus/Response Sequences 15
4.2.3 Functional Requirements 16
4.3 Chemical Interaction Knowledge Graph & Model Comparison 16
4.3.1 Description and Priority 16
4.3.2 Stimulus/Response Sequences 17
4.3.3 Functional Requirements 17
4.4 Personalized Risk Scoring & Explainability 17
4.4.1 Description and Priority 17
4.4.2 Stimulus/Response Sequences 18

4.4.3 Functional Requirements 18
4.5 Scan History & Estimated Dietary Exposure Tracking 18
4.5.1 Description and Priority 18
4.5.2 Stimulus/Response Sequences 18
4.5.3 Functional Requirements 19
4.6 Safer-Alternative Recommendation 19
4.6.1 Description and Priority 19
4.6.2 Stimulus/Response Sequences 19
4.6.3 Functional Requirements 19
4.7 Pediatrician Portal (Secondary Prototype) 19
4.7.1 Description and Priority 19
4.7.2 Stimulus/Response Sequences 20
4.7.3 Functional Requirements 20
4.8 Regulatory Dashboard (Secondary Prototype) 20
4.8.1 Description and Priority 20
4.8.2 Stimulus/Response Sequences 20
4.8.3 Functional Requirements 20
4.9 Parent Authentication & Account Management 21
4.9.1 Description and Priority 21
4.9.2 Stimulus/Response Sequences 21
4.9.3 Functional Requirements 21
5. Other Nonfunctional Requirements 22
5.1 Performance Requirements 22
5.2 Safety Requirements 22
5.3 Security Requirements 22
5.4 Software Quality Attributes 23
5.5 Business Rules 23
6. Other Requirements 24
Appendix A. Glossary 25
Appendix B. Analysis Models 27
B.1 Sample of the Curated Additive-Interaction Dataset 31
B.2 Traceability – Proposal Objectives to System Features 32
Appendix C. To Be Determined List 33

Revision History

Name	Date	Reason for Changes	Version
P.K.R. Abeysekara	26-Sep-2026	Initial SRS draft based on the approved project proposal	1.0

List of Figures

Figure 1: NutriSafe AI – System Architecture Overview 5
Figure 2: Home/Scan screens (Consumer Mobile Application) 10
Figure 3: Child Profile screen (Consumer Mobile Application) 10
Figure 4: Scan Result screen (Consumer Mobile Application) 11
Figure 5: Explanation screen (Consumer Mobile Application) 11
Figure 6: Use Case Diagram 27
Figure 7: Data Flow Diagram (Level 1) 28
Figure 8: Entity Relationship Diagram 29
Figure 9: Class Diagram (Application / Service Layer) 30
Figure 10: State Transition Diagram – Scan Session Lifecycle 31

List of Tables

Table 1: User classes and characteristics 6
Table 2: Software components 12
Table 3: Acronyms and abbreviations 25
Table 4: Sample curated additive-interaction records 31
Table 5: Traceability – proposal objectives to system features 32

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) is an outline of the functional and non-functional requirements for version 1.0 of the NutriSafe AI, which is a multi-modal and explainable AI system designed to assess risks and chemical interaction in terms of personalized pediatric food additive exposure. This SRS includes the overall software system, consumer mobile app (primary deliverable), pediatrician and regulatory web dashboards (secondary deliverables), and backend AI pipelines.

1.2 Document Conventions

Functional requirements are tagged with a unique identifier in the format REQ-<Feature Number>.<Sequence> (ex: REQ-1.1). Unless a requirement explicitly states otherwise, all requirements within a System Feature inherit the priority stated for that feature in Section 4. The keyword “TBD” indicates that information is not yet final, all such items are tracked in Appendix C.

1.3 Intended Audience and Reading Suggestions

This SRS is intended for the following readers:

- Module Supervisor / Examiners – to assess the completeness and correctness of the requirements against the approved proposal.
- The Developer (the author, acting as sole developer) – to guide system design, implementation and testing.
- Prospective Stakeholders (parents, and clinician/regulator reviewers) – to verify that their needs have been correctly captured.

Readers are encouraged to read the context and scope (Sections 1 and 2) and the interface expectations (Section 3) first, then proceed to the functional requirements (Section 4) and non-functional requirements (Section 5). Reviewers may prefer to go directly to Sections 4 and 5, referring back to Section 2 for context as needed.

1.4 Product Scope

NutriSafe AI is a multi-modal, explainable AI system that uses a smartphone camera to capture food ingredient labels and extract ingredient information using OCR/vision-language processing. As an alternative input method, the system can scan a product barcode or QR code to identify the product and retrieve its ingredient information from an external food-product database, where such information is available. The retrieved or extracted ingredients are then assessed against a personalized child health profile. The system moves beyond generic ingredient look-up by considering evidence-supported additive interactions, allergen/health-condition relevance, and an estimated dietary exposure index over time. This solves the problem of ordinary parents being unable to interpret chemical names and E-Numbers on packaging, and of existing food-scanning applications (such as Yuka [1] and Open Food Facts [2]) providing only generic, non-personalized product information.

The primary deliverable is a fully implemented consumer mobile application for parents. A shared backend also supports functional prototype interfaces for pediatricians and food-safety authorities, demonstrating- without requiring live institutional integration- how the same evidence could support clinical and regulatory use in future. The core research contribution evaluated by this system is whether a Relational Graph Convolutional Network (R-GCN) can improve additive-interaction screening compared with a rule-based baseline, while still producing an explainable, personalized result. NutriSafe AI is explicitly a screening and decision-support tool: it does not diagnose medical conditions, measure biological chemical concentrations, or replace professional pediatric medical advice.

1.5 References

[1] "Yuka - The Mobile App That Scans Your Products," Yuka. Accessed: Aug. 20, 2026. [Online]. Available: https://yuka.io/en/

[2] "https://world.openfoodfacts.org." Accessed: Aug. 20, 2026. [Online]. Available: https://world.openfoodfacts.org

[3] M. Zitnik, M. Agrawal, and J. Leskovec, "Modeling polypharmacy side effects with graph convolutional networks," *Bioinformatics*, vol. 34, no. 13, pp. i457–i466, Jul. 2018, doi: 10.1093/bioinformatics/bty294.

[4] F. Zhan and S. Lu, “ESIR: End-to-end Scene Text Recognition via Iterative Image Rectification,” Apr. 02, 2019, *arXiv*: arXiv:1812.05824. doi: 10.48550/arXiv.1812.05824.

[5] S. Lundberg and S.-I. Lee, “A Unified Approach to Interpreting Model Predictions,” Nov. 25, 2017, *arXiv*: arXiv:1705.07874. doi: 10.48550/arXiv.1705.07874.

[6] D. McCann *et al.*, “Food additives and hyperactive behaviour in 3-year-old and 8/9-year-old children in the community: a randomised, double-blinded, placebo-controlled trial,” *Lancet*, vol. 370, no. 9598, pp. 1560–1567, Nov. 2007, doi: 10.1016/S0140-6736(07)61306-3.

2. Overall Description

2.1 Product Perspective

NutriSafe AI is a new, self-contained system it is not a follow on to an existing product and does not replace any existing food database. The system consists of four main logical components:

(1) A consumer mobile application used by parents to scan labels, manage a child profile and view results.

(2) A secondary Clinical Pediatrician Portal prototype.

(3) A secondary Regulatory Dashboard prototype.

(4) A shared backend AI pipeline and REST API that serves all three client interfaces.

The overall architecture is a client-server design consisting of a React Native/Flutter mobile client, React.js web dashboards, a Python/FastAPI backend exposing the AI pipeline, a Neo4j chemical knowledge graph, and a PostgreSQL relational store. Figure 1 shows the major components of the pipeline and how they interconnect.

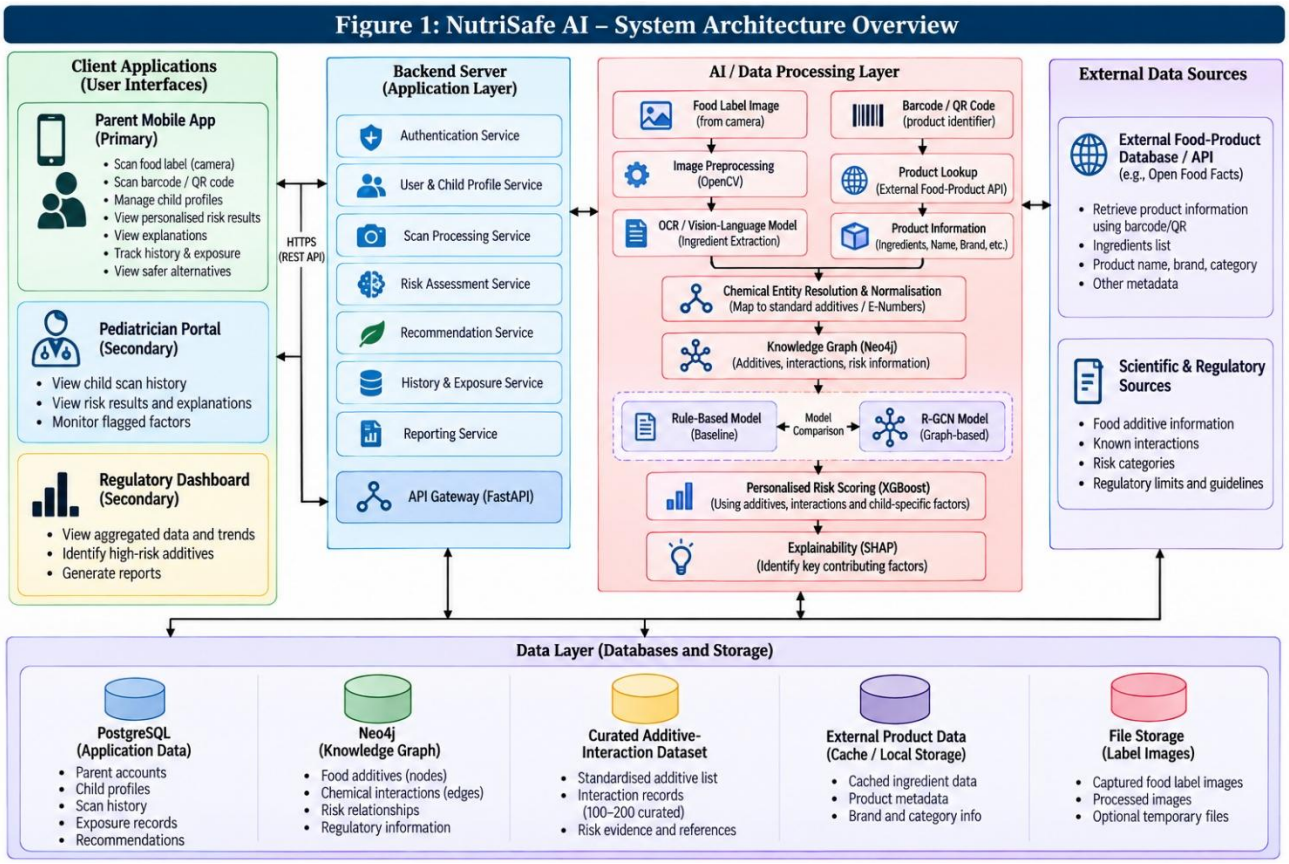


Figure 1: NutriSafe AI – System Architecture Overview

2.2 Product Functions

At a high level, the system will let users:

- Capture a food ingredient label using a smartphone camera and extract its ingredient text using OCR/vision-language processing.
- Scan a product barcode or QR code to identify the product and retrieve available ingredient information from an external food-product database.
- If barcode-based ingredient information is unavailable or incomplete, allow the parent to scan the physical ingredient label instead.
- Normalize the retrieved or OCR-extracted ingredient information into standardised chemical additive identifiers.
- Create and maintain one or more child health profiles (age, allergies, health conditions, food sensitivities).

- Assess detected additives for evidence-supported interactions using a rule-based baseline and a Relational Graph Convolutional Network (R-GCN), compared directly.
- Receive a personalized, explainable risk result (Low / Moderate / High Concern, 0–100 score) with the contributing factors shown.
- View a scan history and an estimated dietary exposure index over a selected period.
- (Pediatrician) View a linked child's logged products and exposure trends as a decision-support view.
- (Regulator) View aggregated, de-identified trends of frequently flagged additives across demonstration data.

2.3 User Classes and Characteristics

Table 1: User classes and characteristics

User Class	Characteristics	Importance
Parent (Primary User)	A parent or guardian scanning food products for a specific child. Assumed to have general smartphone literacy but no specialist knowledge of chemistry or nutrition science.	Primary user class; the system's core value proposition is built around this user.
Pediatrician (Secondary User)	A healthcare professional with a parent-granted link to a specific child's profile, using the portal for decision support rather than as a diagnostic tool.	Secondary but important — demonstrates the clinical extension of the same evidence.
Food-Safety Regulator (Secondary User)	A food-safety authority reviewer exploring aggregated, de-identified market-level trends via the demonstration dashboard.	Secondary — demonstrates the regulatory extension of the same evidence, without live institutional integration.

2.4 Operating Environment

- Client (Parent): Android/iOS smartphone with a functioning camera, built with React Native or Flutter.
- Client (Pediatrician/Regulator): A modern desktop or mobile web browser (Chrome, Edge, Firefox, Safari) with JavaScript and internet access; built with React.js and Tailwind CSS.
- Backend: Python 3 with the FastAPI framework, serving a RESTful JSON API and hosting the vision/OCR, chemical entity resolution, interaction-modelling, risk-scoring and explainability services.
- Knowledge Graph: Neo4j, storing additives, evidence-supported interaction relationships and health-effect references.
- Relational Data: PostgreSQL, storing user accounts, child profiles, scan history and application data.
- Hosting: A cloud platform capable of running a Python/FastAPI application, a Neo4j instance and a PostgreSQL database; exact provider TBD (Appendix C).

2.5 Design and Implementation Constraints

- The system must be implemented using the approved technology stack: React Native/Flutter, React.js + Tailwind CSS, Python + FastAPI, Neo4j, PostgreSQL, PyTorch Geometric (R-GCN), XGBoost and SHAP. [3]
- The chemical-interaction knowledge graph must be built from a literature-curated, evidence-supported dataset of approximately 100–200 interaction records, not an automatically mined large-scale dataset, given the 8-month academic timeline.
- The system must not perform clinical diagnosis, measure actual biological/physiological chemical concentration, or present its output as a medical determination.
- The Pediatrician Portal and Regulatory Dashboard must be implemented as functional prototypes using demonstration/synthetic data; the system must not require live hospital EHR or national food-regulator integration.

- The system must not include in-app e-commerce, direct food purchasing, or live biological/bacteriological laboratory testing functionality.
- Source code must be maintained under version control using Git, with the repository hosted on a platform such as GitHub.

2.6 User Documentation

The following user documentation will be delivered alongside the software: a short on-screen "How it works" guide or tooltip set for first-time parent users, covering scanning, profile creation and interpreting a risk result; and a brief guide for pediatrician and regulator reviewers describing the demonstration data shown in their respective prototypes. No printed manuals will be produced.

2.7 Assumptions and Dependencies

- It is assumed that a bounded, literature-curated dataset of approximately 100–200 evidence-supported additive-interaction records can be compiled within the project timeline from published food-safety and toxicology sources.
- It is assumed that a representative sample of food product label photographs can be collected or sourced for OCR development, testing and evaluation.
- It is assumed that pretrained OCR/vision-language and biomedical NLP components provide sufficient baseline accuracy without requiring the project to train a vision model from scratch.
- It is assumed that synthetic/demonstration data is acceptable for the Pediatrician Portal and Regulatory Dashboard prototypes, since live institutional data is out of scope.
- The project depends on continued access to supervisor consultation for any further requirement validation during SRS review.
- Barcode-based ingredient retrieval depends on the availability, completeness and accuracy of the selected external food-product database. Therefore, barcode lookup shall be treated as a convenience input route rather than the sole source of ingredient information.

3. External Interface Requirements

3.1 User Interfaces

The consumer mobile application includes a clean, card-based interface design with a bottom navigation bar (Home, History, Profile, More). The Home/Scan page has two distinct functions – “Scan Food Label” and “Scan Barcode.” The function of the “Scan Food Label” involves scanning of the physical ingredient label and extraction of the ingredient text via OCR/vision-language processing. The “Scan Barcode” function does not extract ingredients directly from the barcode. Instead, it reads the product identifier and uses it to query an external food-product database for available product and ingredient information. If no reliable ingredient information is available for the scanned product, the application shall prompt the user to scan the physical ingredient label. The Child Profile screen lists the child's name, age, allergies, sensitivities and health conditions in an editable, sectioned form. The Scan Result screen presents a color-coded risk category and 0–100 score prominently at the top, followed by a “Contributing Factors” list. The Explanation screen presents SHAP-derived risk factors as labelled progress bars with a plain-language summary beneath. The interface must be responsive and usable on typical Android/iOS phone screen sizes, since parents are expected to use the system while shopping. The Pediatrician Portal and Regulatory Dashboard will be visually distinguished from the parent app (for example, by a different color accent and a login-protected route) and will present logged/aggregated data as charts and tables rather than a scan-first layout. Consistent navigation, visible loading and error states, and confirmation prompts prior to destructive actions (such as deleting a child profile) are standard UI conventions followed throughout.

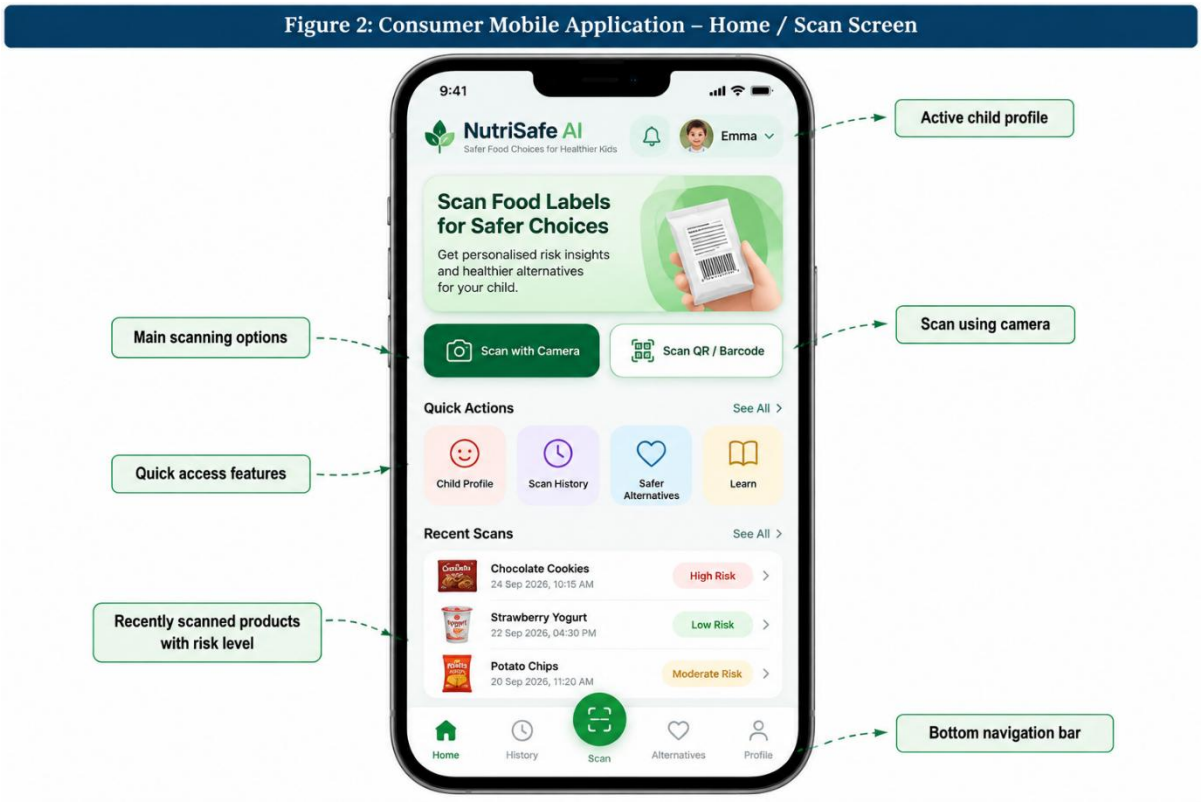


Figure 2: Home/Scan screens (Consumer Mobile Application)

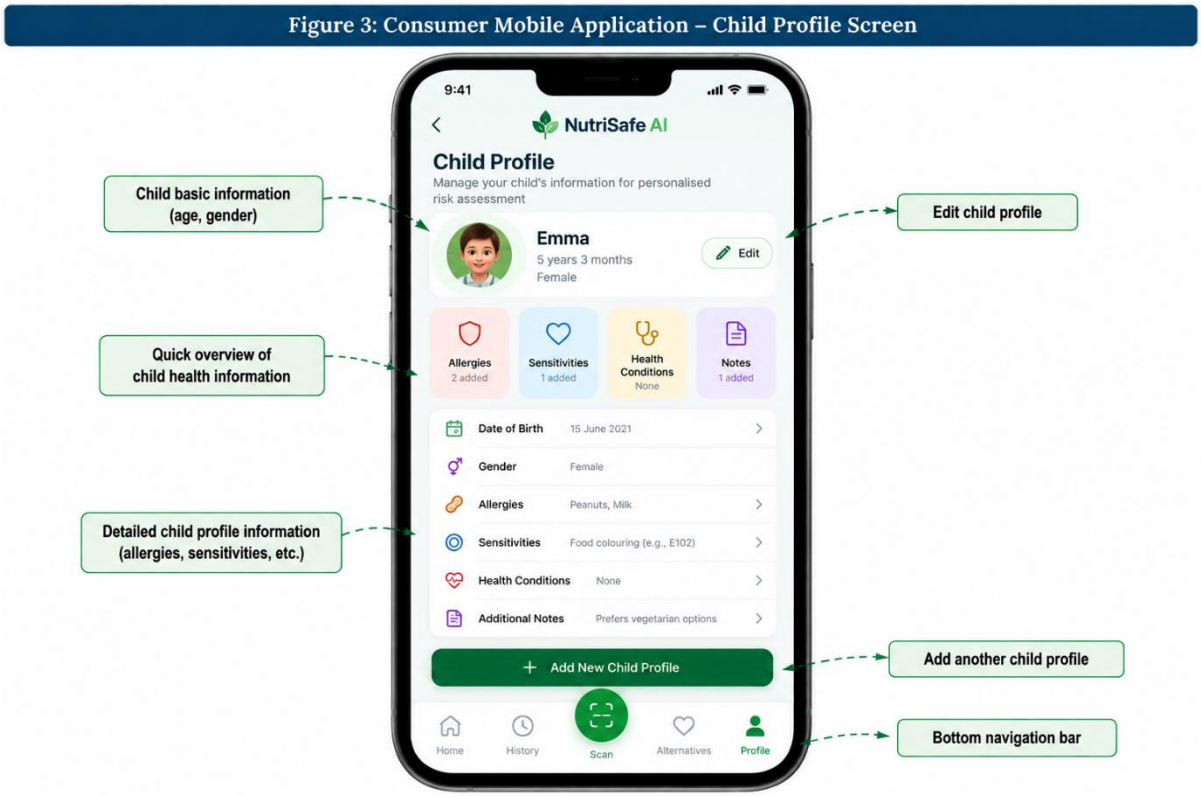


Figure 3: Child Profile screen (Consumer Mobile Application)

Figure 4: Consumer Mobile Application – Scan Result Screen

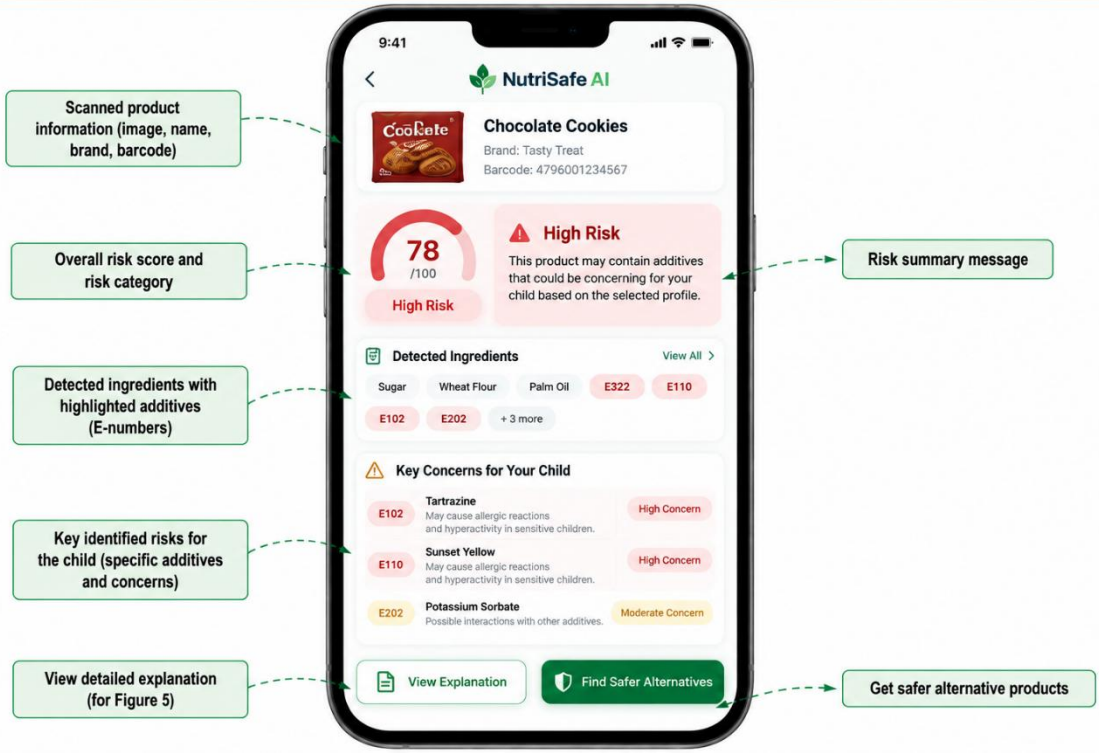


Figure 4: Scan Result screen (Consumer Mobile Application)

Figure 5: Consumer Mobile Application – Explanation Screen

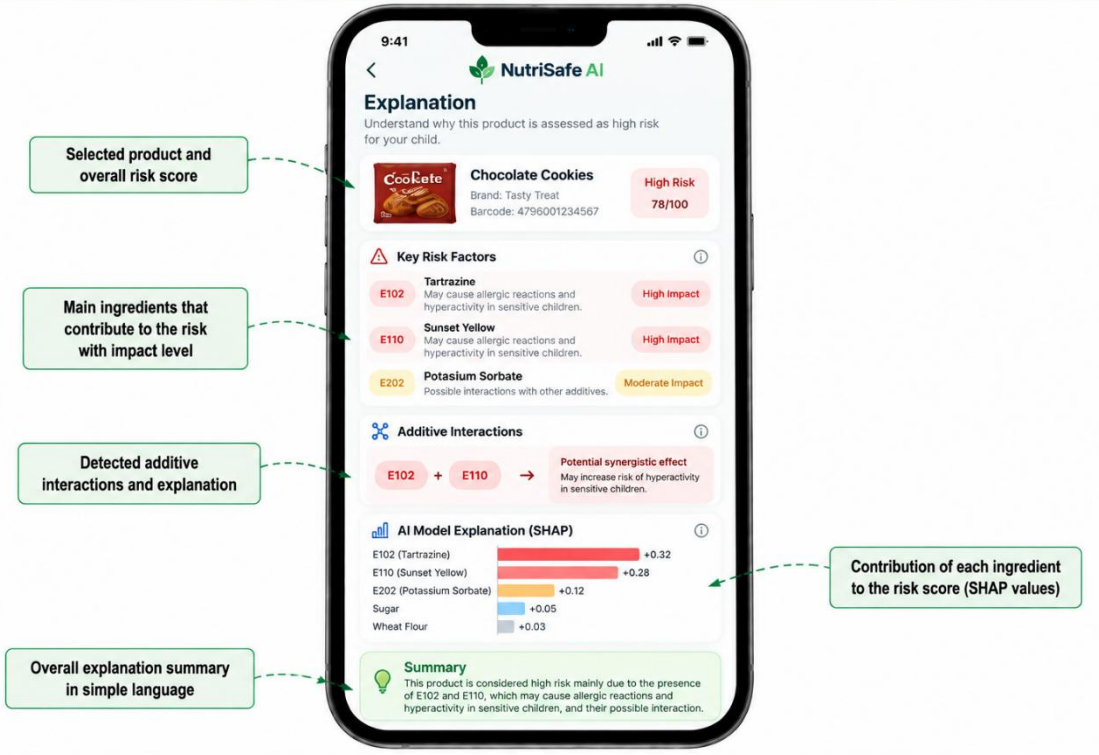


Figure 5:Explanation screen (Consumer Mobile Application)

3.2 Hardware Interfaces

The system has no direct hardware interfaces beyond the client device's standard components. It relies on the parent's smartphone camera for label/QR-barcode capture, and on the device's standard touchscreen and network hardware for interaction and connectivity. No specialised sensors, wearables or laboratory hardware are required or supported in this version.

3.3 Software Interfaces

Table 2: Software components

Component	**Purpose / Interaction**
Neo4j	Primary store for the chemical knowledge graph (additives, interaction evidence, health-effect references). Accessed by the backend's Knowledge Graph Service.
PostgreSQL	Relational store for parent accounts, child profiles, scan records and application data. Accessed by the backend via repository services.
FastAPI REST API	Exposes JSON-based REST endpoints consumed by the mobile app and web dashboards for scanning, profile management, risk retrieval and dashboard aggregation.
OCR / Vision-Language Service	Third-party or open-source OCR/VLM engine used to extract ingredient text from photographed labels.
Chemical Dictionary / Biomedical NLP	Curated reference dictionary (and biomedical NLP model where required) used to normalise ingredient names and E-Numbers to standard additive identifiers.
Web Browser	Renders the React.js single-page applications for the Pediatrician Portal and Regulatory Dashboard.
External Food-Product Database / API	Receives a scanned barcode/QR product identifier and returns available product metadata and ingredient information. The system

	shall treat this as an external data source and fall back to food-label OCR when reliable ingredient information is unavailable.

3.4 Communications Interfaces

JSON formatted REST API requests and responses are used for all communication between the FastAPI backend and the mobile/web clients over HTTP/HTTPS. HTTPS (TLS) must be used to protect data in transit, particularly parent login credentials and child health data, whenever the system is deployed to a public host. This version does not require email, SMS, or push-notification infrastructure; automated recall-style push notifications are explicitly out of scope (see the approved project proposal, Section 6.4).

4. System Features

Functional requirements are organized by system feature, following the pipeline described in the approved project proposal. Each feature lists its priority, typical stimulus/response sequences, and detailed functional requirements.

4.1 Food Label, QR & Barcode Scanning

4.1.1 Description and Priority

This feature provides two input routes for obtaining food ingredient information. In the primary route, the parent captures a photograph of the physical ingredient label, and the system extracts the ingredient text using OCR/vision-language processing [4]. For the alternative pathway, the parent can scan the barcode/QR code. The purpose of scanning the barcode/QR code is to identify the product using an outside food-product database to obtain ingredient information based on the barcode/QR code; however, the barcode/QR code itself is not considered as an ingredient list. In case of the inability to obtain the ingredient information, the user will be asked to scan the ingredient label. Both routes then pass the obtained ingredient information to the chemical entity-resolution stage.

Priority – High.

4.1.2 Stimulus/Response Sequences

● Parent taps “Scan Food Label” and photographs the ingredient label → system preprocesses the image and extracts raw ingredient text using OCR/vision-language processing.
● Parent taps “Scan Barcode” → system reads the barcode/QR product identifier → system queries the configured external food-product database → if ingredient information is available, the system retrieves it.
● Barcode lookup does not return reliable ingredient information → system informs the parent that ingredient information is unavailable and requests a photograph of the physical ingredient label.
● System obtains ingredient information through either OCR/VLM or product-database lookup → system normalizes ingredient names and E-Numbers into standardized additive identifiers.

4.1.3 Functional Requirements

REQ-1.1: The system shall allow a parent to capture a photograph of a food ingredient label using the device camera.

REQ-1.2: The system shall extract ingredient text from a captured food-label image using OCR/vision-language processing.

REQ-1.3: The system shall allow a parent to scan a product barcode or QR code as an alternative product-identification method.

REQ-1.4: The system shall use the scanned barcode/QR identifier to query a configured external food-product database for available product and ingredient information.

REQ-1.5: The system shall not assume that the barcode itself contains the product's ingredient list.

REQ-1.6: If reliable ingredient information is returned by the external product database, the system shall pass the retrieved ingredient information to the chemical entity-resolution stage.

REQ-1.7: If the barcode/QR lookup fails, the product is not found, or reliable ingredient information is unavailable, the system shall prompt the parent to scan the physical ingredient label using the camera.

REQ-1.8: If ingredient text cannot be extracted from a photographed label with sufficient confidence, the system shall request the parent to retake the photograph rather than proceeding with an empty or unsupported result.

REQ-1.9: The system shall normalize ingredient names and E-Numbers obtained through either input route into standardized chemical additive identifiers using the curated reference dictionary.

4.2 Child Health Profile Management

4.2.1 Description and Priority

Allows a parent to create and maintain one or more child health profiles containing the information needed for personalized risk assessment: age, known allergies, selected health conditions and food sensitivities. Priority – High.

4.2.2 Stimulus/Response Sequences

- Parent selects “Create Child Profile” → system presents a form for name, age, allergies, sensitivities and health conditions.
- Parent edits an existing profile → system saves the updated profile and applies it to future scans.
- Parent switches between multiple child profiles → system displays results relevant to the currently selected child.

4.2.3 Functional Requirements

REQ-2.1: The system shall automatically assign a unique system-generated Child ID to each child profile.

REQ-2.2: Each child profile shall be associated with exactly one parent account through the parent's unique User ID.

REQ-2.3: The system shall allow a parent to create and maintain multiple child profiles under the same parent account.

REQ-2.4: Each child profile shall store the child's display name, date of birth, known allergies, food sensitivities and selected health conditions.

REQ-2.5: The system shall calculate the child's age from the stored date of birth where age is required for personalized risk assessment.

REQ-2.6: The parent shall be able to select an active child profile before performing a personalized food assessment.

REQ-2.7: Each completed scan, risk assessment and exposure-history record shall be associated with the unique Child ID of the selected child.

REQ-2.8: The system shall allow the parent to edit or delete a child profile subject to confirmation before deletion.

4.3 Chemical Interaction Knowledge Graph & Model Comparison

4.3.1 Description and Priority

Represents additives and their evidence-supported relationships in a chemical knowledge graph, and evaluates detected additive combinations using both a rule-based baseline and a Relational Graph Convolutional Network (R-GCN), compared directly. This is the core research feature of the system (see the approved proposal, Sections 3.2 and 6.2).

Priority – High.

4.3.2 Stimulus/Response Sequences

- System receives a set of standardized additive identifiers from Feature 4.1 → system queries the knowledge graph for known relationships among the detected additives.
- System runs the rule-based baseline and the R-GCN model on the same detected combination → system records both outputs for comparison and passes the selected result forward.
- No known interaction evidence exists for a given combination → system proceeds using only single-additive evidence and flags the absence of combination evidence in the explanation.

4.3.3 Functional Requirements

REQ-3.1: The system shall represent additives, their categories and evidence-supported interaction relationships in a Neo4j knowledge graph.

REQ-3.2: The system shall maintain a literature-curated interaction dataset of approximately 100–200 evidence-supported records, each linked to its source.

REQ-3.3: The system shall implement a rule-based baseline that flags known additive-combination interactions present in the curated dataset.

REQ-3.4: The system shall implement an R-GCN model, trained on the curated dataset, to predict potentially relevant additive-combination interactions.

REQ-3.5: The system shall record and allow comparison of the rule-based baseline and R-GCN outputs for the same detected combination, to support the evaluation described in Section 6.6 of the approved proposal.

4.4 Personalized Risk Scoring & Explainability

4.4.1 Description and Priority

Combines detected additives, interaction evidence and child-specific factors into a personalized 0–100 risk score and category, and explains the contributing factors using SHAP. [5]

Priority – High.

4.4.2 Stimulus/Response Sequences

- System receives interaction evidence and the active child profile → system computes a personalized risk score using the XGBoost risk model.
- System computes a risk score → system classifies the result into Low / Moderate / High Concern and generates a SHAP-based explanation.
- Parent taps "View Explanation" → system displays the ranked contributing factors behind the result.

4.4.3 Functional Requirements

REQ-4.1: The system shall compute a personalized risk score (0–100) for each scan, combining detected additives, interaction evidence, allergen/health-condition relevance and estimated dietary exposure.

REQ-4.2: The system shall classify each risk score into one of three categories: Low Concern, Moderate Concern, or High Concern.

REQ-4.3: The system shall generate a SHAP-based explanation identifying the factors that contributed most to a given risk result.

REQ-4.4: The system shall present the risk category, score and explanation to the parent without presenting the result as a medical diagnosis (see Section 5.2).

4.5 Scan History & Estimated Dietary Exposure Tracking

4.5.1 Description and Priority

Maintains a history of a child's scanned products and derives an estimated dietary exposure index over a selected period (e.g., the last 7 days), based on recorded scans rather than laboratory measurement. Priority – Medium.

4.5.2 Stimulus/Response Sequences

- Parent completes a scan → system appends the scan and its result to the child's scan history.

- Parent opens “Exposure History” → system displays recent scans, per-product concern levels, and a trend of estimated exposure over the selected period.

4.5.3 Functional Requirements

REQ-5.1: The system shall record each completed scan, including detected additives, risk category and timestamp, against the active child profile.

REQ-5.2: The system shall display a chronological scan history for a selected child.

REQ-5.3: The system shall compute and display an estimated dietary exposure index over a selectable period (like last 7 days), clearly labelled as an estimate rather than a measured biological quantity.

4.6 Safer-Alternative Recommendation

4.6.1 Description and Priority

Where suitable product data is available, suggests safer alternative products when a scanned item is classified as Moderate or High Concern. This is a secondary, lower-priority feature relative to the core risk-assessment pipeline (see the approved proposal, Section 6.4). Priority – Low.

4.6.2 Stimulus/Response Sequences

- System classifies a product as Moderate or High Concern → system searches the product reference data for lower-risk alternatives in the same category using semantic similarity (FAISS).

4.6.3 Functional Requirements

REQ-6.1: Where suitable reference product data is available, the system shall suggest at least one lower-risk alternative product for a Moderate or High Concern result.

REQ-6.2: The absence of a suitable alternative shall not block or delay display of the primary risk result.

4.7 Pediatrician Portal (Secondary Prototype)

4.7.1 Description and Priority

Provides a pediatrician with a parent-granted, read-focused view of a child's logged products and estimated exposure trends, to support — not replace — clinical judgement. Implemented as a functional prototype using demonstration data (see the approved proposal, Section 6.1). Priority – Medium.

4.7.2 Stimulus/Response Sequences

- Pediatrician logs into the portal with a linked child → system displays the child's scan and exposure history as charts.
- Pediatrician selects a flagged product → system displays the factors behind that flag.

4.7.3 Functional Requirements

REQ-7.1: The system shall allow a pediatrician with a valid, parent-granted link to view a specific child's logged scan and exposure history.

REQ-7.2: The system shall present the factors behind a flagged product to the pediatrician in the same explainable format shown to the parent.

REQ-7.3: The system shall present all pediatrician-facing output as decision-support information and shall not present it as an automated clinical decision or diagnosis.

4.8 Regulatory Dashboard (Secondary Prototype)

4.8.1 Description and Priority

Provides a food-safety regulator with an aggregated, de-identified view of frequently flagged additives and market-level trends across demonstration data. Implemented as a functional prototype and explicitly does not integrate with a live national regulatory system (see the approved proposal, Section 6.4). Priority – Medium.

4.8.2 Stimulus/Response Sequences

- Regulator opens the dashboard → system displays aggregated counts of the most frequently flagged additives across demonstration data.
- Regulator filters by category or period → system updates the aggregated view accordingly.

4.8.3 Functional Requirements

REQ-8.1: The system shall display, to an authenticated regulator user, aggregated and de-identified counts of the most frequently flagged additives across demonstration scan data.

REQ-8.2: The system shall not expose any parent- or child-identifiable data through the Regulatory Dashboard.

4.9 Parent Authentication & Account Management

4.9.1 Description and Priority

Preserves the privacy of child health data by limiting access to a parent's own account, and provides basic account management. Priority – Medium.

4.9.2 Stimulus/Response Sequences

- Parent registers with an email and password → system creates an account and requires authentication for subsequent access.
- Parent logs in → system verifies credentials and grants access to that parent's own child profiles and scan history only.

4.9.3 Functional Requirements

REQ-9.1: The system shall require a parent to authenticate with an email/username and password before accessing child profiles or scan history.

REQ-9.2: The system shall store parent passwords using a secure, one-way hashing mechanism; passwords shall never be stored or transmitted in plain text.

REQ-9.3: The system shall restrict a parent's access to only the child profiles and scan history associated with that parent's own account.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

- The Scan Result screen shall display a personalized risk result within approximately 5 seconds of image capture under normal mobile network and processing conditions, for the curated interaction dataset size described in Section 2.5.
- The knowledge-graph interaction query (rule-based and R-GCN comparison) shall complete within approximately 2 seconds for a single scan's detected additive set.
- The Regulatory Dashboard's aggregated view shall render within approximately 3 seconds for the demonstration dataset.

5.2 Safety Requirements

NutriSafe AI is designed for food-safety screening and decision support only. The system shall not present its output as a medical diagnosis, shall not claim to measure the actual biological or physiological concentration of a chemical in a child's body, and shall not replace professional pediatric medical advice. Where a result is Moderate or High Concern, the system shall recommend that the parent consult a qualified healthcare professional. This disclaimer shall be visible within the application and is consistent with the Feasibility, Academic Contribution, Limitations & Evaluation section of the approved project proposal (Section 6.5).

5.3 Security Requirements

- Parent access to child profiles, scan history and exposure data shall require authentication (see REQ-9.1–REQ-9.3).
- All data transmitted between client and server shall be encrypted using HTTPS/TLS whenever the system is deployed to a public environment.
- Pediatrician access to a child's data shall require an explicit, parent-granted link; a pediatrician shall not be able to browse child profiles they have not been linked to.
- The Regulatory Dashboard shall only ever expose aggregated, de-identified data, never individual parent- or child-identifiable records.

- Child health data (allergies, conditions) shall be treated as sensitive data and stored with access limited to the owning parent account and any explicitly linked pediatrician.

5.4 Software Quality Attributes

- Usability: The parent app must be easily useable by first-timers without the need for any external help, as per the usability provided by the card-based interface shown in Figures 2-5.
- Explainability: Every risk result shall be accompanied by a SHAP-based explanation; the system shall not present an unexplained risk label.
- Maintainability: The codebase shall follow a clear separation between the mobile/web clients, the FastAPI backend services (vision/OCR, entity resolution, knowledge graph, interaction modelling, risk scoring, explainability) and the data layer, mirroring the class structure in Figure 8.
- Portability: The parent-facing mobile app shall function on current Android and iOS versions; the web dashboards shall function on the latest versions of major browsers.
- Reliability: If the interaction-modelling service is temporarily unavailable, the system shall fall back to the rule-based baseline rather than failing to return a result.

5.5 Business Rules

- Only an authenticated parent may create, edit or delete a child profile linked to their own account; a pediatrician may only read data for a child they have been explicitly linked to.
- An additive-interaction record may only enter the knowledge graph if it is linked to a cited, published evidence source, consistent with the curated-dataset approach described in the approved proposal.
- A risk result must always be accompanied by its category (Low/Moderate/High Concern), its 0–100 score, and at least one contributing factor; a bare numeric score without explanation shall not be displayed.
- Regulatory Dashboard views must be built only from de-identified, aggregated data; no individual scan record may be directly exposed to a regulator user.

6. Other Requirements

Internationalization: Version 1.0 of the system will be presented in English only; Sinhala and Tamil language support is not required for this version but may be considered as a future enhancement. Legal: NutriSafe AI is not a certified medical device and makes no claim to be one; no specific medical-device regulatory certification applies to this version, consistent with its explicit framing as a screening and decision-support tool (see Section 5.2). Standard good-practice data-protection principles apply to the storage of child health data. Reuse: Where feasible, the Pediatrician Portal and Regulatory Dashboard should share common UI building blocks and the same backend REST API used by the mobile application, to reduce duplication of effort within the project timeline.

Appendix A. Glossary

Table 3: Acronyms and abbreviations

Term	Definition
Additive	A chemical substance (e.g., a preservative, colourant or flavour enhancer) added to a food product, typically identified by a name or E-Number.
E-Number	A code (e.g., E211) used in food labelling to identify an approved food additive.
Chemical Knowledge Graph	A Neo4j graph representing additives and their evidence-supported relationships, health effects and interaction evidence.
R-GCN	Relational Graph Convolutional Network; the graph neural network model compared against the rule-based baseline for interaction screening.
Rule-Based Baseline	A deterministic lookup approach that flags known additive-combination interactions present in the curated dataset, used as the comparison point for the R-GCN model.
SHAP	SHapley Additive exPlanations; a model-agnostic explainability technique used to attribute a risk score to its contributing input factors.
XGBoost	The gradient-boosted decision-tree model used to compute the personalised 0–100 risk score.
OCR / VLM	Optical Character Recognition / Vision-Language Model; used to extract ingredient text from a photographed food label.
Risk Category	The classification of a scan result into Low, Moderate or High Concern.
Estimated Dietary Exposure Index	A derived measure of a child's recorded exposure to a given additive over a selected period, based on logged scans rather than laboratory measurement.
Child Profile	The record of a child's age, allergies, health conditions and food sensitivities used to personalise a risk assessment.

Secondary Prototype	A functional but intentionally lightweight implementation (Pediatrician Portal, Regulatory Dashboard) built to demonstrate a use case without full institutional integration.
SRS	Software Requirements Specification.
REST API	Representational State Transfer Application Programming Interface; the interface exposed by the FastAPI backend.
FYP	Final Year Project.
Barcode / Product Identifier	A machine-readable code used to identify a packaged food product. In NutriSafe AI, the identifier is used to query an external product database; it is not assumed to contain the ingredient list itself.
External Food-Product Database	A third-party or externally maintained data source that can map a product identifier to available product metadata and ingredient information.

Appendix B. Analysis Models

This appendix presents the preliminary analysis models developed during requirements analysis: a Level-1 Data Flow Diagram, an Entity Relationship Diagram, a Class Diagram of the application/service layer, and a State Transition Diagram for the scan-session lifecycle. These models are consistent with, and elaborate on, the six-layer pipeline described in Section 6.2 of the approved project proposal.

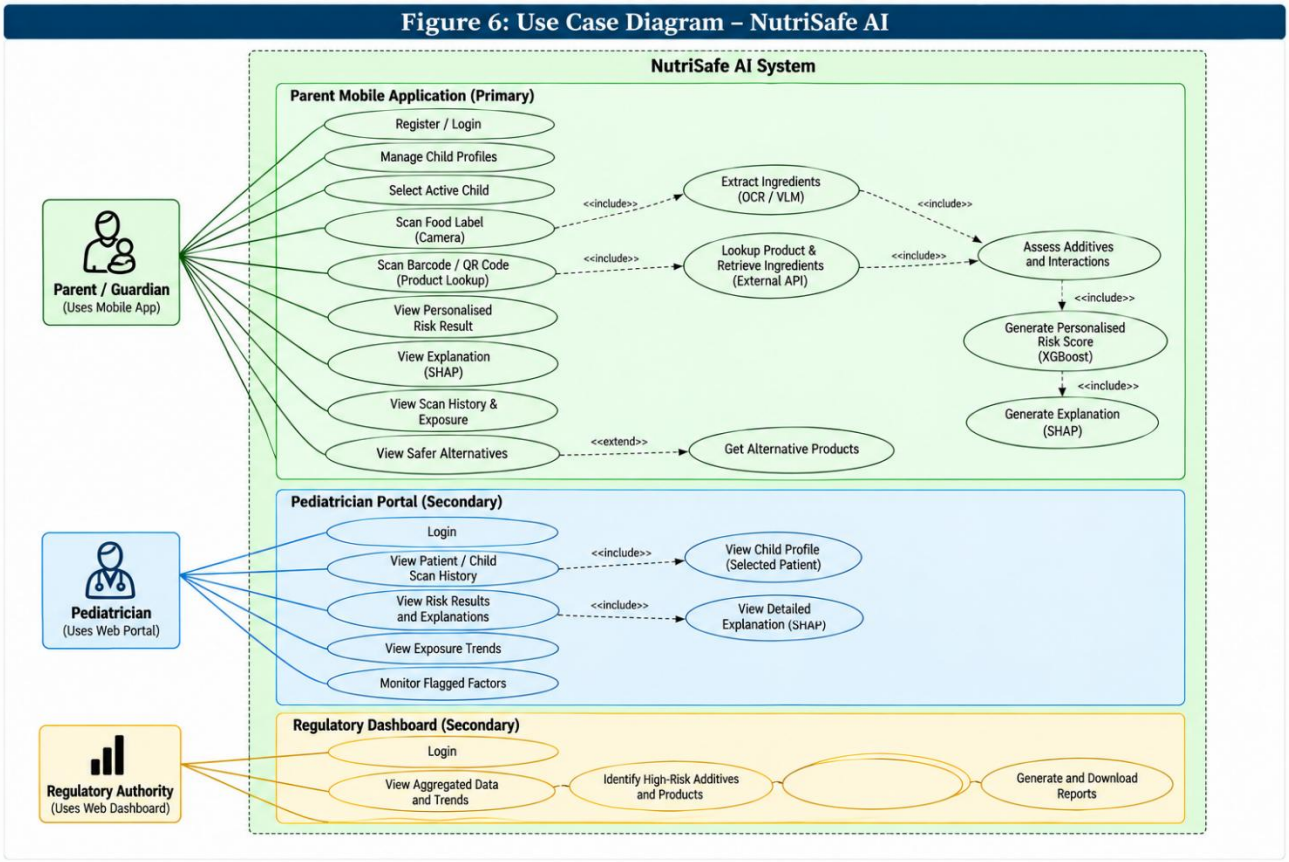


Figure 6: Use Case Diagram

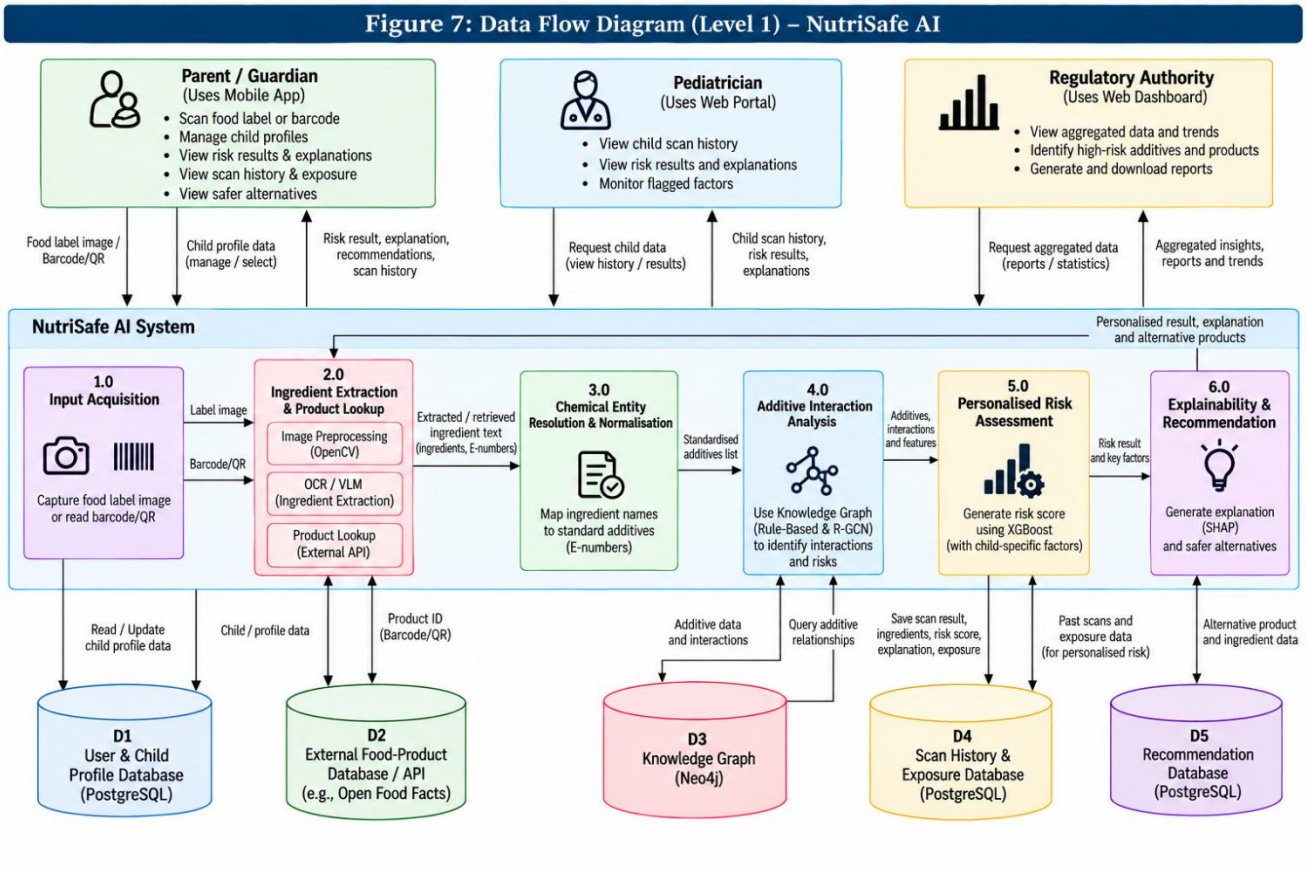


Figure 7: Data Flow Diagram (Level 1)

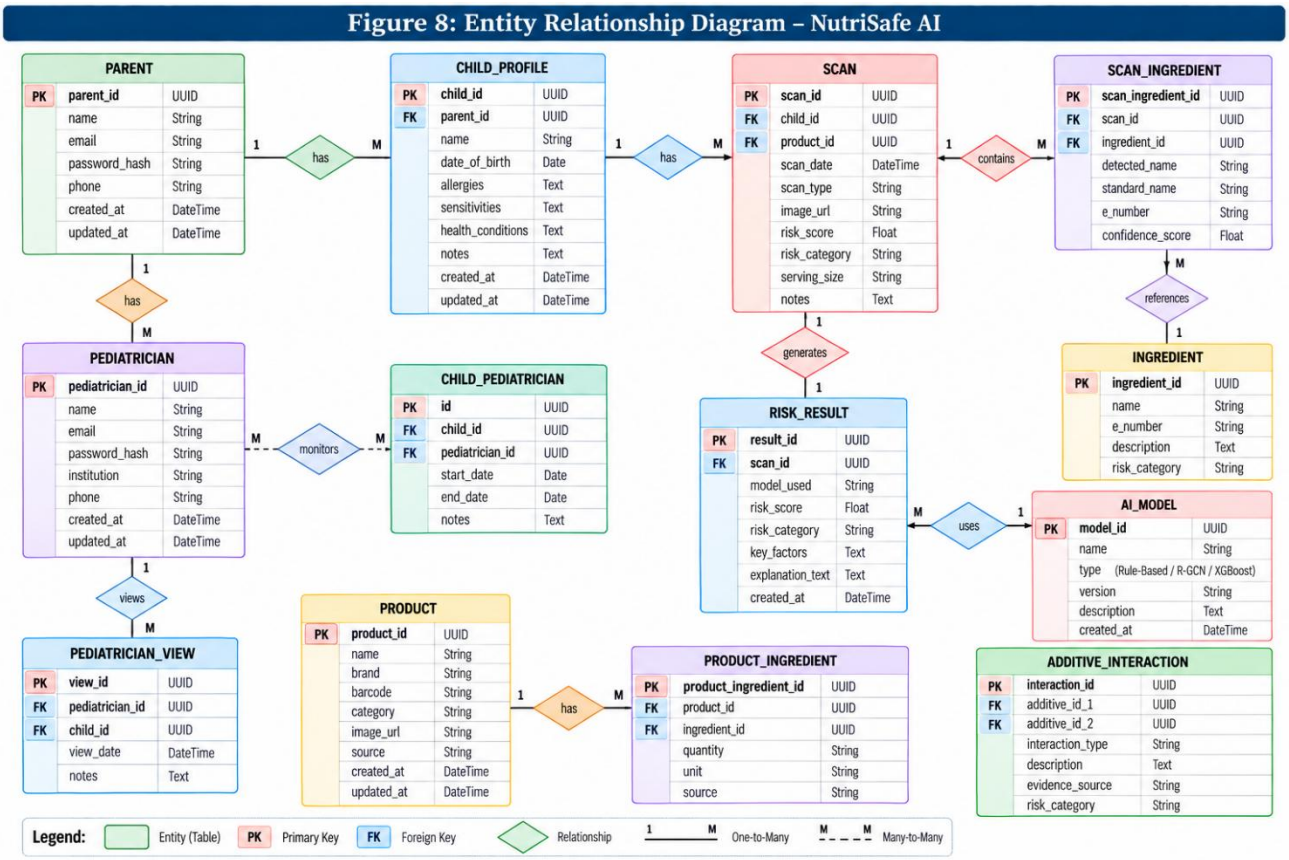


Figure 8: Entity Relationship Diagram

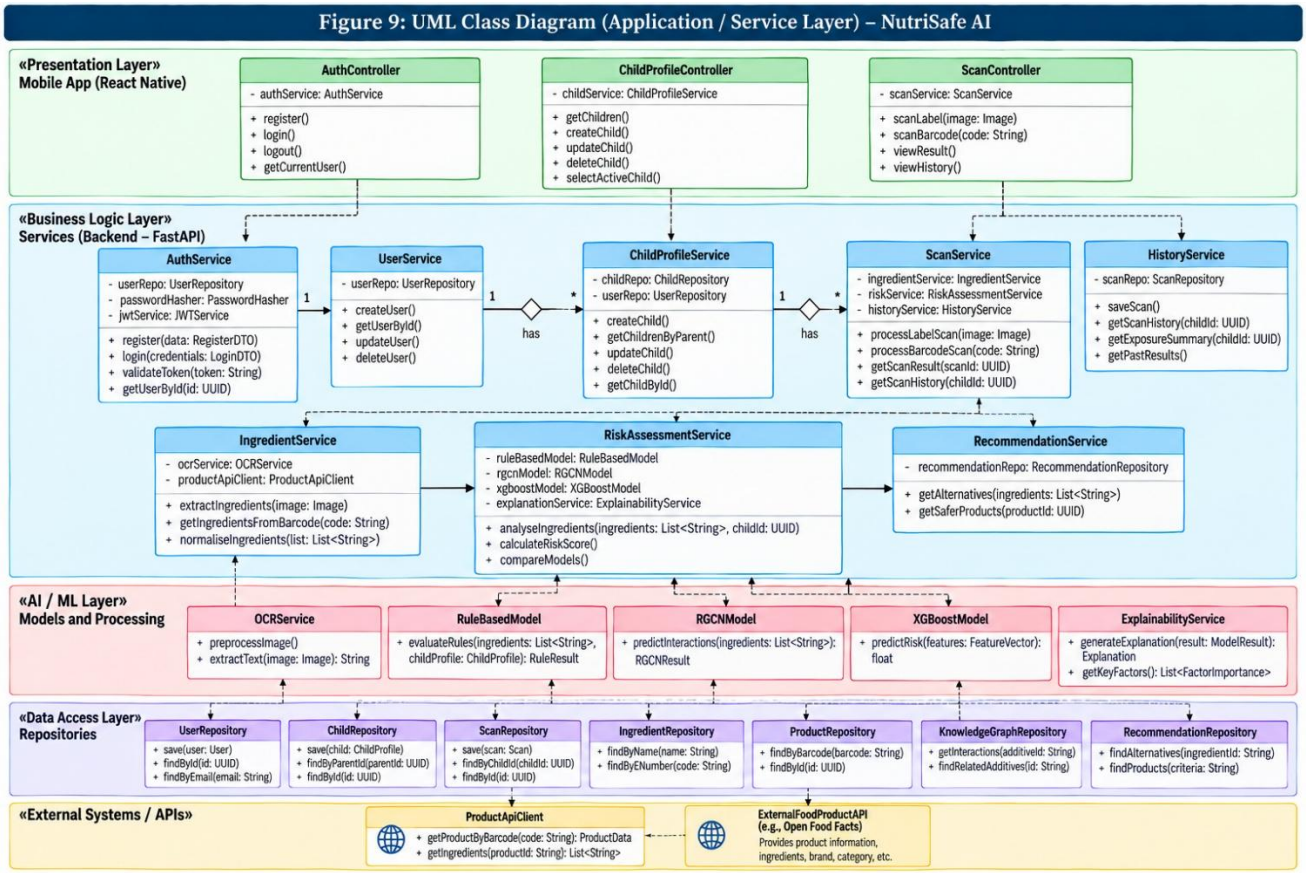


Figure 9: Class Diagram (Application / Service Layer)

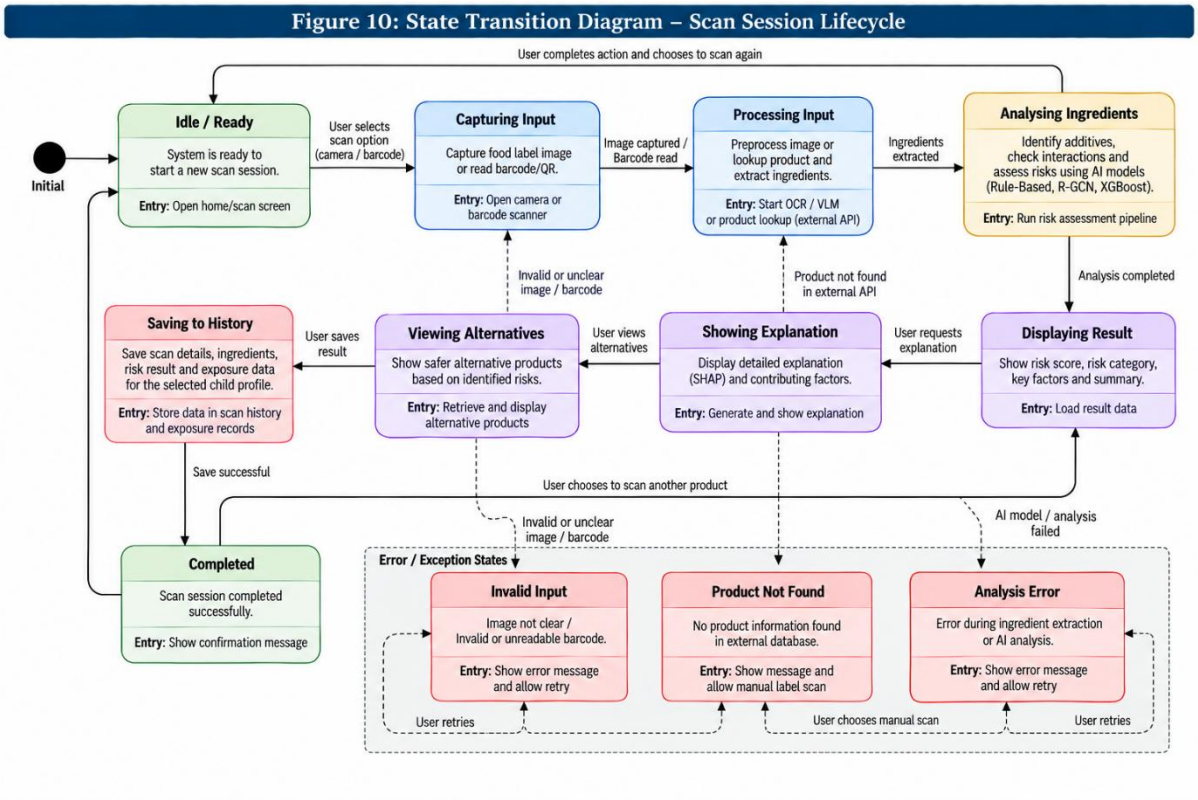


Figure 10: State Transition Diagram – Scan Session Lifecycle

B.1 Sample of the Curated Additive-Interaction Dataset

The table below illustrates the structure and level of detail expected for the literature-curated interaction dataset described in Sections 2.5 and 4.3; the final dataset will contain approximately 100–200 such records, each linked to a cited source.

Table 4: Sample curated additive-interaction records

Additive A	Additive B	Interaction Note	Evidence Source
Sodium Benzoate (E211)	Sunset Yellow FCF (E110)	Combination associated with increased hyperactive behavior in young children in a randomized controlled trial.	McCann et al., 2007 [6]
Monosodium Glutamate (E621)	Tartrazine (E102)	Illustrative combination flagged for further evidence	TBD – literature review in progress

		review during dataset curation.	
Sodium Nitrite (E250)	Ascorbic Acid (E300)	Illustrative combination noted in food-chemistry literature; to be verified during dataset curation.	TBD – literature review in progress

B.2 Traceability – Proposal Objectives to System Features

Table 5: Traceability – proposal objectives to system features

Proposal Objective (Section 5)	**System Feature(s)**
Review existing food-scanning apps and chemical-interaction research	Informs Sections 1.4, 2.1 and 4.3 (context, not a system feature)
Build an OCR/vision pipeline and chemical entity resolution	4.1 Food Label, QR & Barcode Scanning
Construct a chemical interaction knowledge graph; compare rule-based baseline with R-GCN	4.3 Chemical Interaction Knowledge Graph & Model Comparison
Develop a personalized, SHAP-explainable risk-scoring model	4.4 Personalized Risk Scoring & Explainability
Implement the mobile app and prototype clinician/regulatory use cases	4.1, 4.2, 4.5, 4.6, 4.7, 4.8, 4.9
Evaluate the system (metrics, extraction accuracy, usability)	Addressed in Section 6.6 of the approved proposal; not a separate system feature

Appendix C. To Be Determined List

This is a dynamic list of open requirements issues that remain to be resolved as the project proceeds:

- Final cloud hosting provider for the backend, Neo4j and PostgreSQL instances (Section 2.4).
- Final choice of OCR/vision-language engine and biomedical NLP component, pending a short evaluation against sample label images (Section 3.3).
- Final external food-product database/API used for barcode-based product and ingredient lookup (for example, Open Food Facts), subject to coverage, API availability, data quality and suitability for the research prototype.
- Exact size and final source list for the curated additive-interaction dataset, pending literature availability (Sections 2.5, 4.3 and Appendix B.1).
- Session/token expiry duration for parent and pediatrician authentication (Section 5.3).
- Whether a minimal safer-alternative product reference dataset is available in time to fully support Feature 4.6, given its Low priority (Section 4.6).
- Exact set of demonstration data used to populate the Pediatrician Portal and Regulatory Dashboard prototypes (Sections 4.7 and 4.8).